GDPA Fasteners
GDPA House, Near Railway Godowns
Jalandhar City-144004 Punjab (INDIA)
T: +91-181-2642001/02/03/04
F: +91-181-2642005
E: exports@gdpaindia.com
W: www.gdpaindia.com

Final Inspection Report

AN ISO 9001:2015 Certified Company
CE 15048 : 2007
CE 14399 : 2015

Doc: QC/QF/04
Rev: 00
Date: 01.08.2018

CUSTOMER NAME	G.W.S. ENGINEERING & INDUSTRIAL SUPPLIES LTD.	INV. NO.	365ES/E-294	INV. DATE	31-01-2026	ORDER NO.	E-214	CERTIFICATE NO.	3652060-H
DESCRIPTION	HEX SCREW(OVERSIZE) ISO 4017 GR-8.8,HEX NUT(OVERTAPP) ISO 4032 CL.10, WASHER ISO 7091 100 HV	SIZE	M20X60	FINISH	HOT DIP GALV.	INV. QTY. IN PCS.	12000	ASSEMBLY DETAIL	ASSEMBLED

HEAD MARKING-BOLT	G,M.20,8.8SB,60	HEAD MARKING-NUT	G 10SB	HEAD MARKING-WASHER	N/A
HEAT NO. BOLT	F255195	HEAT NO. NUT	4523251	HEAT NO. WASHER	T5C501300
BUYER ORDER NO.	PO000002577	BUYER PART NO.	ISO4017G20X060	ASSEMBLY BATCH NUMBER	GD214040126B

CERTIFICATE IN ACCORDANCE WITH EN -10204 TYPE 3.1 (2013)

CHEMICAL COMPOSITION (AS PER 898 PART 1-2013 & 2-2012)

HEX SCREW (ISO 4017)

		C%	Mn%	S%	P%	Si%	Cr%	Mo%	B%
M20X60	SPECIFIED	0.20-0.55		0.025 MAX.	0.025 MAX.				0.003 MAX.
	OBSERVATION	0.295	1.085	0.008	0.022	0.084	0.441		0.003

HEX NUT (ISO 4032)

		C%	Mn%	S%	P%	Si%	Cr%	Mo%	B%
M.20	SPECIFIED	0.58 MAX.	0.30 MIN.	0.058 MAX.	0.048 MAX.				
	OBSERVATION	0.426	0.675	0.043	0.033	0.258	0.524		0.002

WASHER (ISO 7091)

		C%	Mn%	S%	P%	Si%	Cr%	Mo%	B%
M.20	SPECIFIED								
	OBSERVATION	0.150	0.417	0.020	0.022	0.232			

MECHANICAL PROPERTIES (AS PER 898 PART 1-2013 & 2-2012)

HEX SCREW (ISO 4017)

	No. of checked Samples		Proof Load (KN) (MIN.)	Suitability test (Bolt, Nut assemblies) (MIN.)	Ultimate test (MPa) (MIN.)	Tensile Strength (N/MM2) (MIN.)	Elongation after fracture % (MIN.)	Reduction Area % (MIN.)	Hardness (HRC)	Impact Strength at -20°C
M20X60	1	SPECIFIED	147.000	203.000	203.000				23-34	27J
		OBSERVATION	147.650	203.900	235.900				31	118

HEX NUT (ISO 4032)

	No. of checked Samples		Hardness(HRC)	Proof Load (KN)
M.20	1	SPECIFIED	26-36	259.700
		OBSERVATION	33	260.150

WASHER (ISO 7091)

	No. of checked Samples		HARDNESS(HV)
M.20	1	SPECIFIED	100
		OBSERVATION	126

DIMENSIONS

HEX SCREW (ISO 4017)

M20X60		No. of checked Pcs	Width across flat (s)	Head Height (K)	Shank Dia (ds)	Thread Length (b)	THREAD DIA.	Total Length (L)	THREAD DIMENSIONS OK WITH NUT
SPECIFIED	MIN	5	29.67	12.28	N/A	N/A	19.62	59.40	OK/NOT OK
	MAX		30.00	12.71			19.95	60.60	
OBSERVATION1		1	29.75	12.40			19.78	60.35	OK
OBSERVATION2		2	29.70	12.35			19.75	60.20	OK
OBSERVATION3		3	29.88	12.55			19.70	59.80	OK
OBSERVATION4		4	29.85	12.50			19.85	59.65	OK
OBSERVATION5		5	29.81	12.56			19.92	60.40	OK

HEX NUT (ISO 4032)

M.20		No. of checked Pcs	Across Flat (S)	Height (m)	Minor Dia.	THREAD DIMENSION OK WITH BOLT
SPECIFIED	MIN	5	29.16	16.90	17.69	OK/NOT OK
	MAX		30.00	18.00	18.14	
OBSERVATION1		1	29.75	17.56	18.02	OK
OBSERVATION2		2	29.65	17.45	17.95	OK
OBSERVATION3		3	29.60	17.80	17.82	OK
OBSERVATION4		4	29.78	17.75	17.80	OK
OBSERVATION5		5	29.80	17.38	18.03	OK

WASHER (ISO 7091)

M.20		No. of checked Pcs	Hole Dia.(d1)	Outer Dia.(d2)	Thickness (h)	Dimensions
SPECIFIED	MIN	5	22.00	35.40	2.40	OK/NOT OK
	MAX		22.52	37.00	3.60	
OBSERVATION1		1	22.35	36.56	3.05	OK
OBSERVATION2		2	22.15	36.50	3.25	OK
OBSERVATION3		3	22.25	36.40	3.20	OK
OBSERVATION4		4	22.50	36.80	2.80	OK
OBSERVATION5		5	22.42	36.75	2.55	OK

COATING (AS PER ISO 10684 /ISO 4042)

COATING (Hot Dip Galv./Zinc/Plain)	No. of samples/specified	Average coating thickness of HDG in microns
M20X60	5PCS./40 µ MIN.	70-100 µ
M.20	5PCS./40 µ MIN.	70-100 µ
M.20	5PCS./40 µ MIN.	65-100 µ

RESULT: Satisfactory

Declaration - This is to certify That,No Russian steel isused to manufacture this material.

GDPA FASTENERS
WORKS AUTHORIZED INSPECTOR
EMANUEL MASIH
JALANDHAR

UK CA
O879

CE
0045-CPR-1536

INSPECTED BY

QA-ACCEPTED

This certificate relates to the tested sample only.

AUTHENTICATION : THIS CERTIFICATE IS AN ORIGINAL ONLY IF SIGNED